Andhra University College of Engineering

Chemical Engineering | Course Code: CH301

Chemical Reaction Engineering - Chemical Reaction Kinetics & Ideal Reactor

---

Overview: Design equations for CSTR in series, plug flow reactor sizing, recycle reactors, non-

I. UNIT-WISE SYLLABUS BREAKDOWN & HIGH-YIELD TOPICS

Unit 1: Homogeneous Reaction Kinetics
Rate equations, Temperature dependency, Arrhenius law, Collision theory, Transition state theory, Ki

Unit 2: Ideal Reactor Design Equations
Material balance for Batch, Continuous Stirred Tank CSTR, and Plug Flow Reactors PFR under constant/

Unit 3: Multiple Reactor Networks & Yield
CSTRs in series and parallel, Autocatalytic reactions, Product selectivity in parallel and series re

Unit 4: Non-Ideal Flow & Residence Time Distribution
RTD functions Et and Ft, Pulse and step tracer experiments, Segregation and dispersion flow models

II. HIGH-YIELD SEMESTER EXAMINATION QUESTIONS

1. Derive the performance equation for an isothermal Plug Flow Reactor PFR for a second-order liquid phas
2. Compare the volume requirements of a single CSTR with two equal-sized CSTRs in series for 90% conversi
3. Explain how residence time distribution RTD is obtained using step tracer stimulus-response experiment

III. PRESCRIBED TEXTBOOKS & ACADEMIC REPOSITORIES

* H. Scott Fogler - Elements of Chemical Reaction Engineering Prentice Hall 5th Ed
* Octave Levenspiel - Chemical Reaction Engineering Wiley 3rd Ed

Verified NoteVault Institutional Archive - Authorized for Registered Student Use Only